

2. Simulation and Verification Results

Four DUT instances were instantiated with different parameter configs for positive weight behavior, negative weight behavior, maximum signed weight, and minimum signed weight. The testbench verifies both functional datapath behavior and AER handshake behavior.

DUT Configurations

DUT	Weight	Threshold	Reset	Primary purpose
dut_pos	+1	500	0	Normal positive update, threshold crossing, spike generation
dut_neg	-1	500	0	Signed negative arithmetic
dut_max	+127	2047	-123	Maximum signed weight/upper bound spike case
dut_min	-128	2047	0	Minimum signed weight/negative saturation

2.1 Reset and Idle Behavior

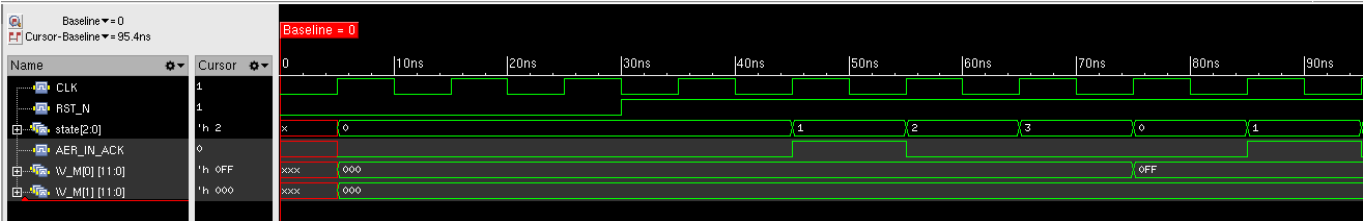
Purpose: Reset must place the block in a deterministic idle condition before any AER transaction begins.

Stimulus: Assert RST_N low, then release reset without applying an input event.

Expected behavior: The FSM is in IDLE, AER_IN_ACK and AER_OUT_REQ are low, representative neuron states are zero, and no unintended transaction begins.

Important Signals:

Signal	Expected behavior	Why it matters
CLK	Free running clock	Provides timing reference for reset release and FSM behavior.
RST_N	Low during reset, then rises	Shows when reset is applied and released.
state	IDLE after reset	Confirms controller returns to the idle state.
AER_IN_ACK	0	No input transaction should be acknowledged.
AER_OUT_REQ	0	No output spike should be requested.
V_M[0]	0	Representative neuron state is cleared.
V_M[1]	0	Second representative neuron state is also cleared.



2.2 Normal Positive Weight Neuron Update

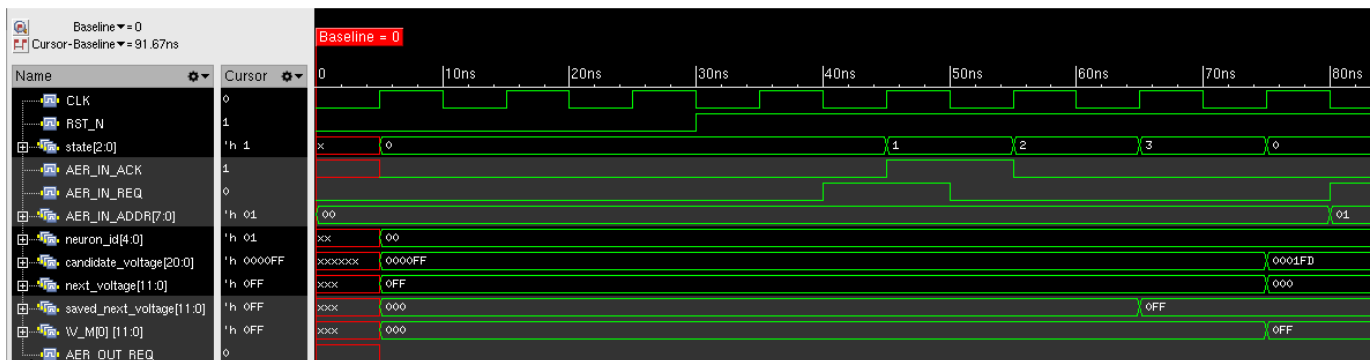
Purpose: A normal AER event must be accepted, processed through the shared datapath, written back to the selected neuron, and completed without generating a spike when the threshold is not reached.

Stimulus: Send a positive weight event to a neuron initially at 0 using dut_pos.

Expected behaviour: The input is acknowledged, the selected neuron is processed, candidate/next voltage evaluate to +255, V_M updates from 0 to 255, and AER_OUT_REQ remains low because $255 < 500$.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_REQ	Pulses high for the input request	Marks the incoming AER transaction.
AER_IN_ADDR	Shows the selected neuron address	Confirms which neuron is targeted.
AER_IN_ACK	Pulses high after acceptance	Confirms the request was accepted.
state	0 -> 1 -> 2 -> 3 -> 0	Shows the normal non spiking FSM path.
neuron_id	Matches the requested neuron	Confirms correct temporary address to neuron selection.
candidate_voltage	+255	Shows the combinational datapath result.
next_voltage	+255	Shows the value chosen for write back.
saved_next_voltage	+255	Shows the transaction result captured by the FSM.
V_M[0]	0 -> 255	Confirms state write back.
AER_OUT_REQ	Remains 0	No spike should be generated below threshold.



2.3 All 32 Neurons Are Updateable

Purpose: Every entry in the 32 neuron state array must be individually selectable and writable under the current temporary neuron selection convention.

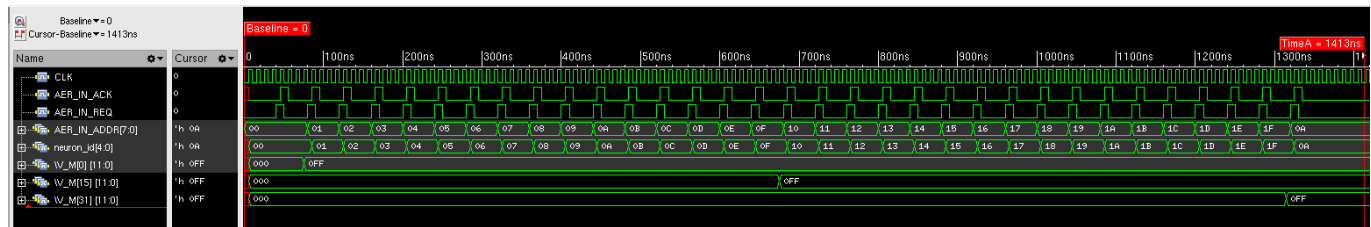
Stimulus: Sequentially apply requests with AER_IN_ADDR values 0x00 through 0x1F.

Expected behaviour: Each address is acknowledged and the corresponding V_M entry is updated. Representative first, middle, and last neurons should visibly change from 0 to 255.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_REQ	Repeated request pulses	Shows a sequence of independent transactions.
AER_IN_ADDR	00 -> 01 -> ... -> 1F	Shows all 32 address values are exercised.
AER_IN_ACK	One ACK per accepted transaction	Confirms each request is accepted.

neuron_id	Tracks addresses 0-31	Confirms temporary selection logic reaches every neuron.
V_M[0]	0 -> 255	Representative first neuron.
V_M[15]	0 -> 255	Representative middle neuron.
V_M[31]	0 -> 255	Representative final neuron.



2.4 Independent State Between Different Neurons

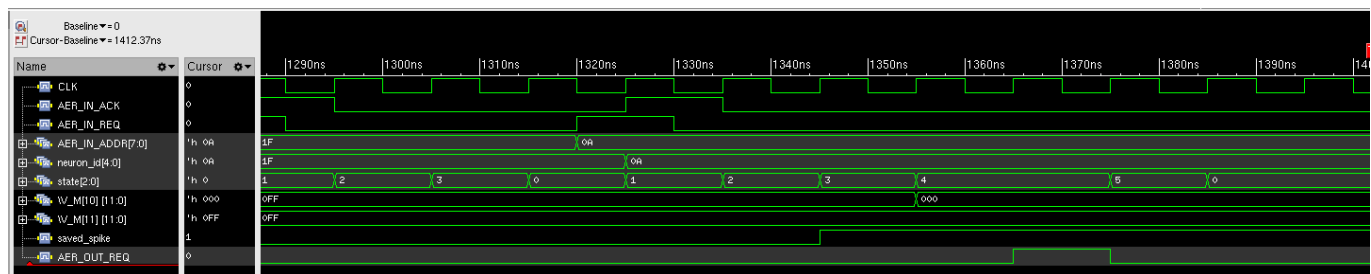
Purpose: Updating one neuron must not corrupt the stored membrane voltage of another neuron.

Stimulus: Start neurons 10 and 11 at 255, then issue another event only to neuron 10.

Expected behaviour: Neuron 10 crosses threshold and resets to 0 while neuron 11 remains at 255.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_ADDR	0x0A	Targets neuron 10.
AER_IN_REQ	Pulses high	Starts the second update.
AER_IN_ACK	Pulses high	Confirms the event is accepted.
state	Processes neuron 10 through the FSM	Shows control sequencing.
neuron_id	10	Confirms the intended neuron is active.
V_M[10]	255 -> 0	Target neuron is updated/reset after spike.
V_M[11]	Remains 255	Non-target neuron is preserved.
saved_spike	Becomes 1	Confirms the neuron 10 transaction crossed threshold.
AER_OUT_REQ	Asserts	Spike generated for neuron 10.



2.5 Threshold Detection and Post Spike Reset

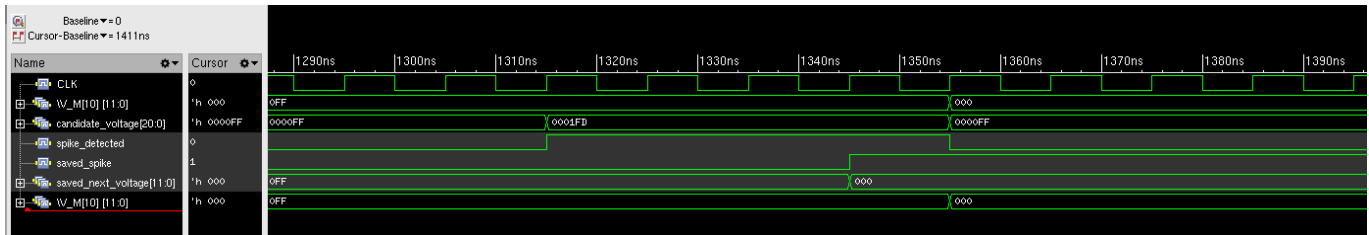
Purpose: The LIF datapath must detect when the updated membrane voltage reaches or exceeds threshold and replace the candidate value with RESET_VALUE.

Stimulus: Apply a second positive event to a neuron already at 255 using threshold 500 and reset 0.

Expected behaviour: The candidate is approximately 509, spike_detected asserts, saved_spike captures 1, saved_next_voltage becomes 0, and V_M is written with 0.

Important Signals:

Signal	Expected behavior	Why it matters
candidate_voltage	Approximately 509	Shows the pre threshold arithmetic result.
spike_detected	1 while candidate >= threshold	Live combinational threshold comparison.
saved_spike	1	Latched spike decision for the transaction.
saved_next_voltage	0	Latched reset value to be written.
V_M[10]	255 -> 0	Confirms post spike reset is committed.



2.6 Outgoing Spike and AER Address Generation

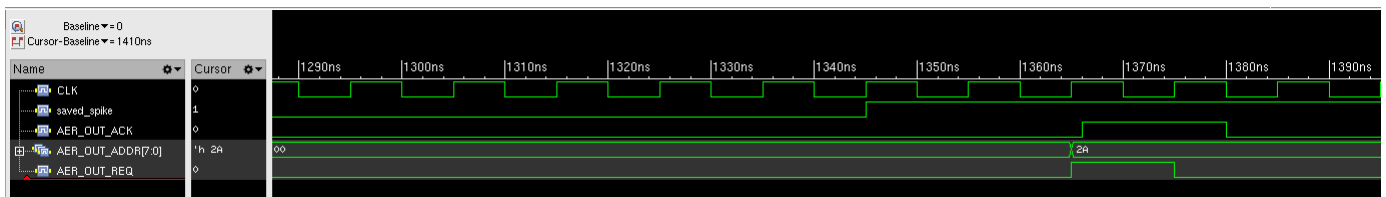
Purpose: A threshold crossing must produce an outgoing AER request with the expected Team-2 address encoding.

Stimulus: Use the neuron #10 threshold crossing from the previous test.

Expected behaviour: AER_OUT_REQ asserts and AER_OUT_ADDR becomes 0x2A: upper nibble 0x2 for Team 2 and lower nibble 0xA for the source neuron under the current interface convention.

Important Signals:

Signal	Expected behavior	Why it matters
saved_spike	1	Indicates that the transaction generated a spike.
AER_OUT_REQ	0 -> 1	Starts the outgoing AER transaction.
AER_OUT_ADDR	0x2A	Confirms expected tile/source encoding.
AER_OUT_ACK	Eventually asserts	Shows the downstream receiver acknowledges the event.



2.7 Negative Weight Arithmetic

Purpose: The signed datapath must correctly handle a negative synaptic weight and update neuron state in the negative direction without falsely generating a spike.

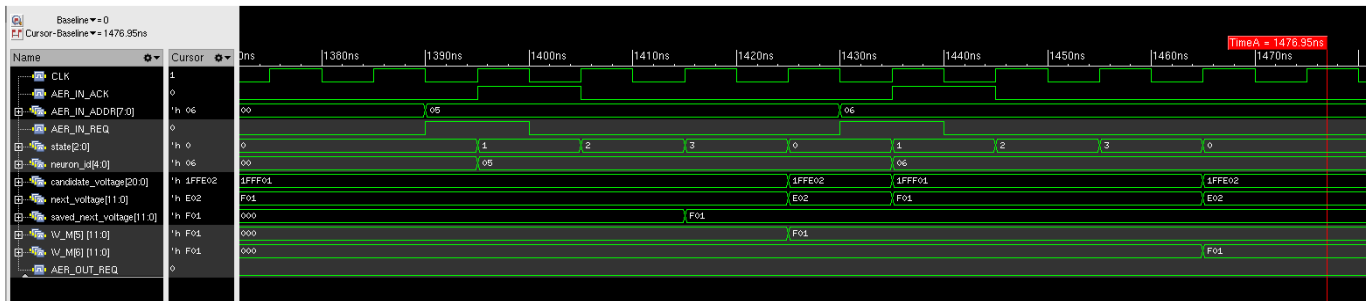
Stimulus: Use dut_neg with W = -1 and send events to neurons 5 and 6.

Expected behaviour: Each selected neuron changes from 0 to -255 and AER_OUT_REQ remains low.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_REQ	Request pulse for each neuron	Starts each negative weight transaction.

AER_IN_ADDR	0x05 then 0x06	Selects neurons 5 and 6.
AER_IN_ACK	ACK for each request	Confirms each transaction is accepted.
state	Normal non-spiking path	Shows controller sequencing.
neuron_id	5 then 6	Confirms target selection.
candidate_voltage	-255	Shows signed negative arithmetic.
next_voltage	-255	Shows selected write back value.
saved_next_voltage	-255	Latched transaction result.
V_M[5]	0 -> -255	First negative update.
V_M[6]	0 -> -255	Second independent negative update.
AER_OUT_REQ	Remains 0	Negative update must not cause a positive threshold spike.



2.8 Input Rejection While Datapath Is Busy

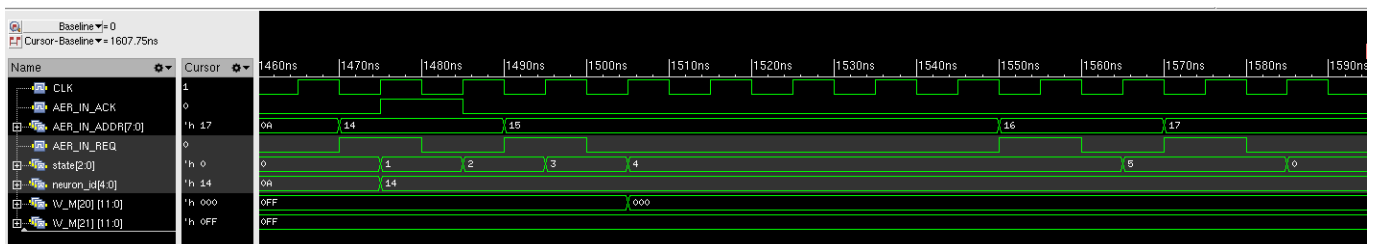
Purpose: Because the tile uses one shared arithmetic datapath, a second input event must not be accepted while another neuron update is still in progress.

Stimulus: Begin processing neuron 20, then assert another request targeting neuron 21 before the first transaction finishes.

Expected behaviour: The second request receives no acknowledgement and V_M[21] remains unchanged while neuron 20 completes.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_REQ	Shows the attempted second request	Indicates new traffic arrives while busy.
AER_IN_ADDR	Changes to neuron 21	Identifies the rejected second target.
AER_IN_ACK	Remains low for the second request	Proves it is not accepted.
state	Non idle while second request is present	Shows the datapath/controller is busy.
neuron_id	Remains associated with active transaction	Shows the current event is preserved.
V_M[20]	Completes its active update	Confirms the original transaction finishes.
V_M[21]	Unchanged	Confirms the rejected request causes no state write.



2.9 Output Backpressure

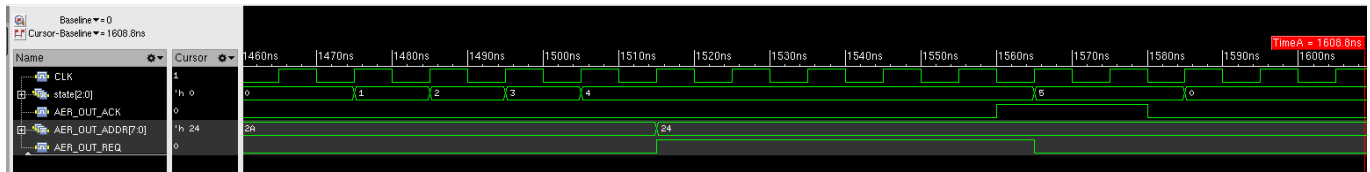
Purpose: The block must preserve an outgoing spike when the downstream receiver is not ready.

Stimulus: Generate a spike and hold AER_OUT_ACK low for several cycles.

Expected behaviour: AER_OUT_REQ remains asserted and AER_OUT_ADDR remains unchanged until ACK is received.

Important Signals:

Signal	Expected behavior	Why it matters
AER_OUT_REQ	Stays high while ACK=0	Preserves the pending outgoing event.
AER_OUT_ADDR	Stable at the spike address	Prevents address corruption under backpressure.
AER_OUT_ACK	Held low, then rises	Models downstream backpressure and eventual acceptance.
state	Remains in output wait state	Shows controller does not complete the transaction prematurely.



2.10 Reject New Input While Output Is Blocked

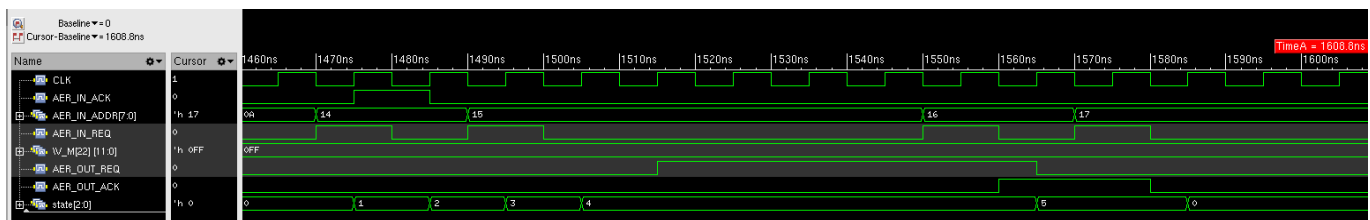
Purpose: A new neuron update must not begin while an outgoing spike transaction is still pending.

Stimulus: While AER_OUT_REQ is high and AER_OUT_ACK is low, apply a new input event for neuron 22.

Expected behaviour: AER_IN_ACK remains low and V_M[22] does not change until the output transaction completes.

Important Signals:

Signal	Expected behavior	Why it matters
AER_IN_REQ	Asserts during blocked output	Creates the attempted new input transaction.
AER_IN_ADDR	Targets neuron 22	Identifies the rejected target.
AER_IN_ACK	Remains 0	Confirms new input is not accepted.
V_M[22]	Unchanged	Confirms no hidden state update occurs.
AER_OUT_REQ	Remains 1	Shows the previous output event is still active.
AER_OUT_ACK	Remains 0 during rejection	Confirms output is still blocked.
state	Remains in output transaction state	Shows controller cannot return to input processing yet.



2.11 Correct Completion of the Output Handshake

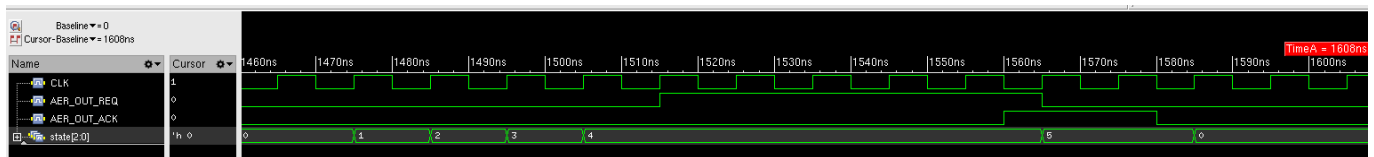
Purpose: The outgoing AER request/acknowledge sequence must complete in the expected four phase order and return the controller to IDLE.

Stimulus: Allow AER_OUT_ACK to assert after a pending AER_OUT_REQ, then release ACK.

Expected behaviour: The sequence is REQ up, ACK up, REQ down, ACK down, followed by state returning to IDLE.

Important Signals:

Signal	Expected behavior	Why it matters
AER_OUT_REQ	Rises first, then falls after ACK	Sender side of the handshake.
AER_OUT_ACK	Rises after REQ, then falls after REQ	Receiver side of the handshake.
state	SEND_SPIKE -> WAIT_OUTPUT_ACK_LOW -> IDLE	Confirms correct control sequencing and recovery.



2.12 Maximum Signed Weight

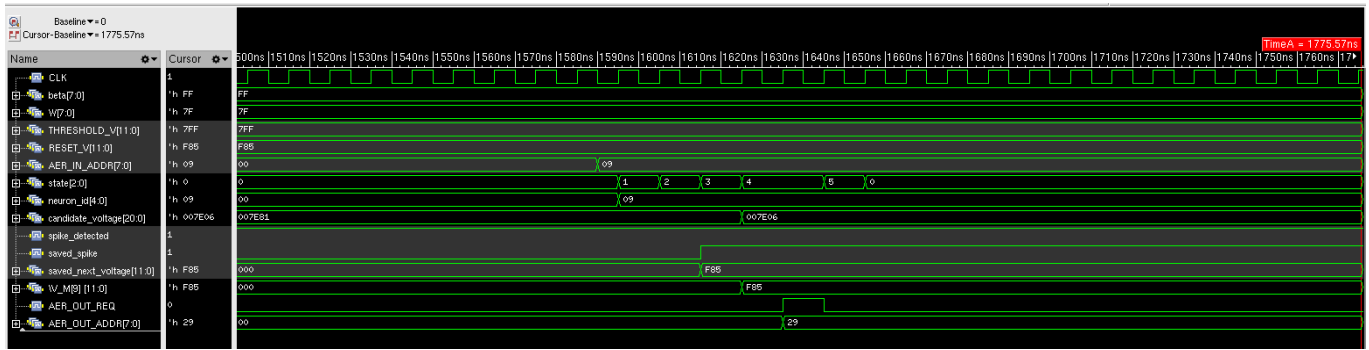
Purpose: The datapath must correctly handle the maximum legal signed 8 bit weight and produce the expected threshold crossing/spike behavior.

Stimulus: Use dut_max with W = +127, THRESHOLD = 2047, RESET = -123 and target neuron 9.

Expected behaviour: The candidate voltage exceeds threshold, the transaction records a spike, V_M[9] is written with -123, and an output event with address 0x29 is generated.

Important Signals:

Signal	Expected behavior	Why it matters
W	+127	Confirms maximum signed positive weight configuration.
THRESHOLD_V	2047	Shows upper configured threshold.
RESET_V	-123	Shows configured post spike reset value.
AER_IN_ADDR	0x09	Targets neuron 9.
state	Spiking FSM path	Shows control transitions through output generation.
neuron_id	9	Confirms target selection.
candidate_voltage	Large positive value > 2047	Shows upper bound arithmetic case.
spike_detected	1	Live threshold result.
saved_spike	1	Latched transaction spike decision.
saved_next_voltage	-123	Latched reset value.
V_M[9]	0 -> -123	Confirms post spike write back.
AER_OUT_REQ	Asserts	Outgoing event generated.
AER_OUT_ADDR	0x29	Expected Team2/source address.



2.13 Minimum Signed Weight and Negative Saturation

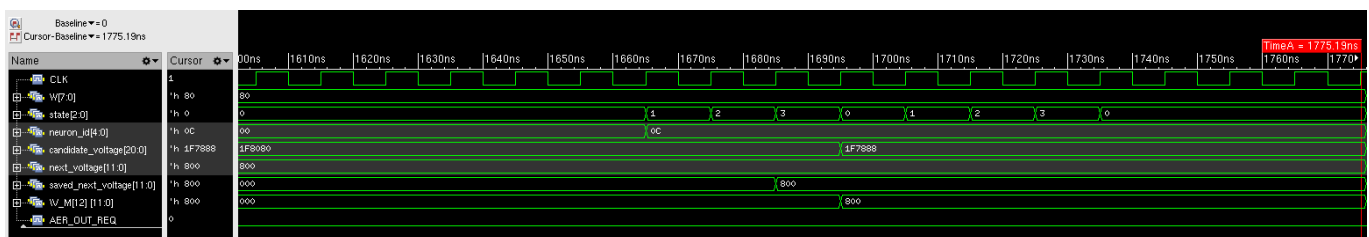
Purpose: The datapath must correctly handle the minimum legal signed 8 bit weight and clamp an out of range negative result instead of wrapping around.

Stimulus: Use dut_min with W = -128 and target neuron 12. Apply the minimum weight update twice.

Expected behaviour: The raw candidate falls below -2048, next_voltage and saved_next_voltage clamp to -2048, V_M[12] becomes -2048, and a second update leaves it at -2048 with no spike.

Important Signals:

Signal	Expected behavior	Why it matters
W	-128	Confirms minimum signed 8 bit weight configuration.
state	Normal non-spiking update path	Shows control flow.
neuron_id	12	Confirms target selection.
candidate_voltage	Below -2048	Shows out of range negative arithmetic result.
next_voltage	-2048	Shows lower saturation/clamp decision.
saved_next_voltage	-2048	Latched saturated write back value.
V_M[12]	0 -> -2048, then remains -2048	Proves saturation without wraparound.
AER_OUT_REQ	Remains 0	Negative saturation must not generate a positive threshold spike.



Verification Summary

Verification point	Expected result	Status
Reset / idle	Known state after reset	PASS
Normal positive update	0 -> 255, no spike	PASS
All 32 neurons	Every neuron entry updateable	PASS
Independent state	Non-target neuron unchanged	PASS
Threshold / reset	Threshold crossing causes reset	PASS
Outgoing spike	REQ/address generated correctly	PASS
Negative weight	0 -> -255, no spike	PASS
Busy input rejection	Second request not accepted	PASS
Output backpressure	REQ/address held stable	PASS
Input blocked by output	New request rejected	PASS

Handshake completion	REQ/ACK sequence returns to IDLE	PASS
Maximum weight	+127 threshold/spike/reset case	PASS
Minimum weight	-128 saturates at -2048	PASS

Testbench summary: (xrun.log uploaded to repo)

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TESTS RUN : 344

TESTS FAILED : 0

RESULT : PASS

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